

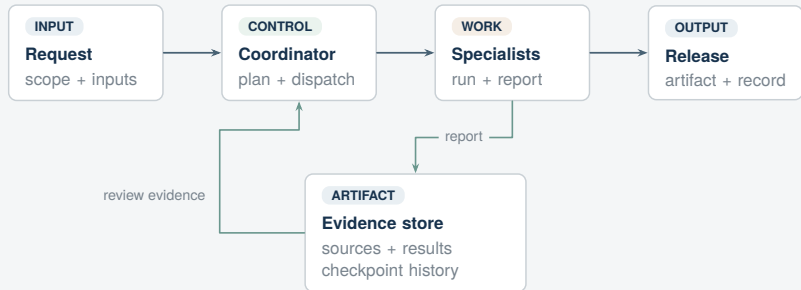
WORKED EXAMPLE · SYSTEM DESIGN

Research system architecture

A six-slide example of overview, branching, feedback, and recovery

Illustrative architecture · September 2026

Coordination produces verified output



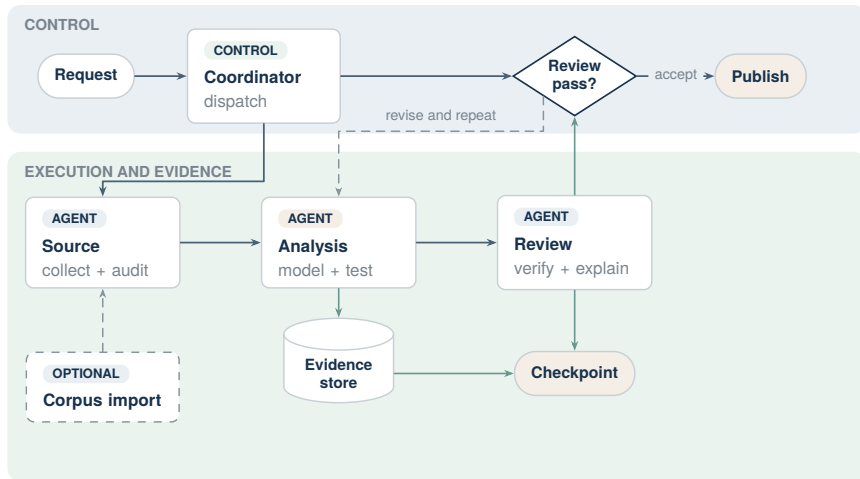
Reading guide

The main path moves left to right.

Evidence returns through a separate lower corridor.

Color encodes role.

Tasks can branch and return to review



Path guide

Control ends after review approval.

Agents run apart from stored artifacts.

- primary
- evidence
- - -> optional / retry

Control owns state; workers return evidence

USER AND EXTERNAL SYSTEMS

Interface

submit request · inspect progress · approve release

CONTROL PLANE

Coordinator

state · policy · scheduling · verification gate

EXECUTION PLANE

AGENT

Source agent

collect + audit

AGENT

Analysis agent

model + test

AGENT

Review agent

check + explain

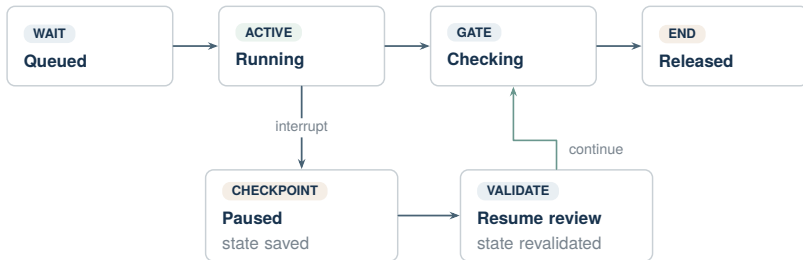
Ownership boundary

Control owns task policy.

Workers own bounded execution and return evidence.

Durable artifacts allow inspection without reopening a running worker.

Resume only after checkpoint validation



Recovery contract

A checkpoint records the completed boundary.

Resume begins with checkpoint validation.

The final transition remains subject to the same verification gate.

State, evidence, and recovery stay separate

SUMMARY 06

CONTROL

The coordinator owns state and release.

It applies policy, schedules bounded work, and opens the verification gate.

EXECUTION

Workers return durable evidence.

Source, analysis, and review agents return artifacts and checkpoints without owning global state.

RECOVERY

Resume begins with validation.

A checkpoint records the completed boundary before execution returns to the verification path.

Paused

Resume review

Checking